Chapter 8

Introduction to Topology

8.1 Why This Matters to Economists

This chapter and the associated quiz 5 on open and closed sets is intended for MRes students and MSc students who wish to take EC487 (Advanced Microeconomics) which is compulsory for MSc Econometrics and Mathematical Economics (EME) students and available to MSc Economics students with permission of the instructor. If you struggle unsuccessfully with the quiz and chapter you are likely to find EC487 very difficult. If you are not an MRes or MSc EME student and have no interest in EC487 do not attempt quiz 5 and read this chapter unless you are very confident that you have a good grasp of the rest of EC400 maths and enjoy the challenge.

This quiz covers the beginnings of real analysis in $\mathcal{R}^n$ and starts the move to topology. This is relatively abstract material. If you have done a course in real analysis this may be useful revision of the basics and might help to build your intuition. The concepts here are required for rigorous proofs of some important results in microeconomics.

The mathematics in this chapter matters to economists for several reasons. There is a big result, that a function that is continuous on a closed and bounded subset of $\mathcal{R}^n$ has a maximum and a minimum. This is important because maximization and minimization crop up very frequently in economics. You will meet the mathematics discussed in this chapter very early in a mathematically rigorous microeconomics course in the form of the continuity assumptions for consumer preferences and its implication, the existence of a continuous utility functions that captures the preferences. Continuity is also central to theorems on the existence of general competitive equilibrium.

Sections 1-4 are very useful. The rest of the chapter is harder going and less widely used in economics. Section 9 is the background you need to understand the material on continuity in consumer theory.

8.2 Vector Length and Open Balls

Remember from the chapter 2 that in $\mathcal{R}^n$ we have a concept of the length or norm of a vector

$$\|\mathbf{x}\| = \left(\sum_{i=1}^{n} x_i^2\right)^{\frac{1}{2}}.$$

In one dimensional space $n = 1$, and $\|x\| = |x|$, the absolute value of x. In two and three dimensional space Pythagoras' Theorem implies that this is indeed the length of the vector. The norm has the property that

$$\|\mathbf{x}\| \geq 0 \text{ for all } \mathbf{x} \tag{8.1}$$

$$\|\mathbf{x}\| = 0 \text{ if and only if } \mathbf{x} = 0. \tag{8.2}$$

The distance between two vectors $\mathbf{x}$ and $\mathbf{x}_0$ in $\mathcal{R}^n$ is defined as

$$\|\mathbf{x} - \mathbf{x}_0\| = \left(\sum_{i=1}^{n} (x_i - x_{0i})^2\right)^{\frac{1}{2}}.$$

In one dimensional space this reduces to the absolute value $|x - x_0|$ which is the distance between the numbers x and x_0. In two and three dimensional space Pythagoras' Theorem implies that $\|\mathbf{x} - \mathbf{x}_0\|$ is the distance between $\mathbf{x}$ and $\mathbf{x}_0$. I therefore refer to $\|\mathbf{x} - \mathbf{x}_0\|$ as distance, even when $n > 3$. Properties 8.1 and 8.2 imply that

$$\begin{aligned} \|\mathbf{x} - \mathbf{x}_0\| &\geq 0 \text{ for all } \mathbf{x} \\ \|\mathbf{x} - \mathbf{x}_0\| &= 0 \text{ if and only if } \mathbf{x} = \mathbf{x}_0. \end{aligned}$$

Now think about the set of points that are closer to $\mathbf{x}_0$ than some strictly positive number δ. In one dimensional space $\mathbf{x}_0$ is a number and this is the open interval $(x_0 - \delta, x_0 + \delta)$. In two dimensional space it is a circle centred on $\mathbf{x}_0$ with radius δ. In three dimensional space this is a sphere or ball. In fact we use the term ball regardless of the dimension of the space, using the following definition.

Definition 48 *If δ is a strictly positive number and $\mathbf{x}_0$ is an element of $\mathcal{R}^n$ the **open ball** $B(\mathbf{x}_0, \delta)$ centred on $\mathbf{x}_0$ with radius δ is the set of points that are closer than δ to $\mathbf{x}_0$, that is*

$$B(\mathbf{x}_0, \delta) = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n, \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

Note that the inequality in this definition $<$ is strict. I have used the word "open" many times before when writing about "open intervals" and I will explain shortly what the word "open" means in general. In order to do this I will introduce some more terms that mathematicians have borrowed from elsewhere.

8.3 Boundaries

Imagine yourself standing on the equator. Take a tiny step to the north; you are in the northern hemisphere. Take a tiny step to the south; you are in the southern hemisphere. The equator is the boundary of the two hemispheres. Mathematicians have borrowed the term boundary from geographers and use it much the same sense, applying it to the n dimensional space of real number $\mathcal{R}^n$.

Definition 49 *A point $\mathbf{x}_0$ in $\mathcal{R}^n$ is an element of the* ***boundary*** *of a subset S of $\mathcal{R}^n$ if for any number $\delta > 0$ the open ball $B(\mathbf{x}_0, \delta)$ contains both points that are elements of S and points that are in the complement S^C of S.*

You may recall that the definition of complement:

Definition 50 *The complement S^C of a subset S of $\mathcal{R}^n$ is the set of elements of $\mathcal{R}^n$ that are not in S, that is*

$$S^C = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n, \, \mathbf{x} \notin S\}.$$

The fact that the radius δ of the open ball $B(\mathbf{x}_0, \boldsymbol{\delta})$ can be a very small positive number captures the idea that a step to the north, however tiny, moves you from the equator to the northern hemisphere. The definitions of the boundary and the complement of a set immediately imply a result that is useful later.

Proposition 51 *The boundary of a set S and the boundary of its complement S^C are the same.*

The idea of a boundary may become clearer with an example.

Example 52 *The boundary of the set*

$$S = \left\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, \, x_1 > x_2\right\}$$

shown in figure 8.1 is the straight line $x_1 = x_2$.

This example is illustrated in Figure 8.1. Any open ball such as A centered on a point on the line $x_1 = x_2$ has points that are in S and points that are not in S, so lies in the boundary of S. However any open ball such as B centered on a point that lies in S does not contain any points that are not in S if its radius is small enough. Similarly an open ball such as C that is centered on a point that does not lie in S or on the line $x_1 = x_2$ does not contain any points that are in S if its radius is small enough. Thus points that are not on the line $x_1 = x_2$ are not on the boundary of S.

8.4 Open and Closed Sets and Boundaries

British suburban gardens usually have something physical marking the boundary, a fence, a hedge or a wall; you could describe them as closed. Suburban gardens in the USA often have nothing visible to mark the boundary; you could describe them as open. Mathematicians use the terms open and closed in a somewhat similar way.

- A subset S of $\mathcal{R}^n$ is **open** if **none** of the points in the boundary of S lie in S.

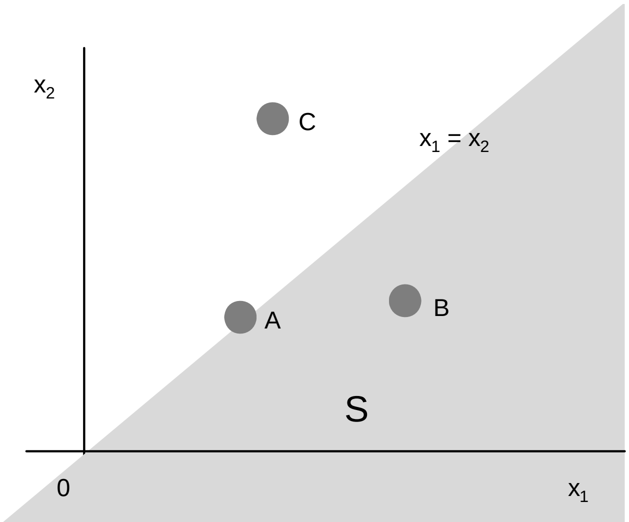

Figure 8.1: The set $\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2,\, x_1 > x_2\}$

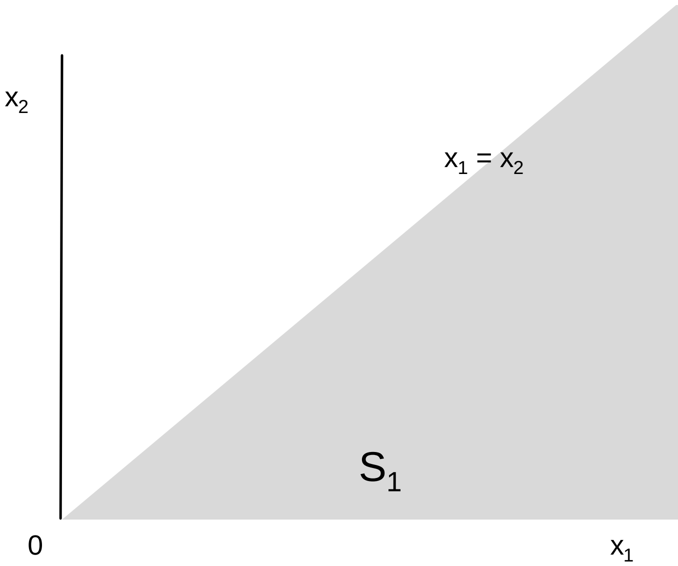

Figure 8.2: $S_1 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 < x_1\}$

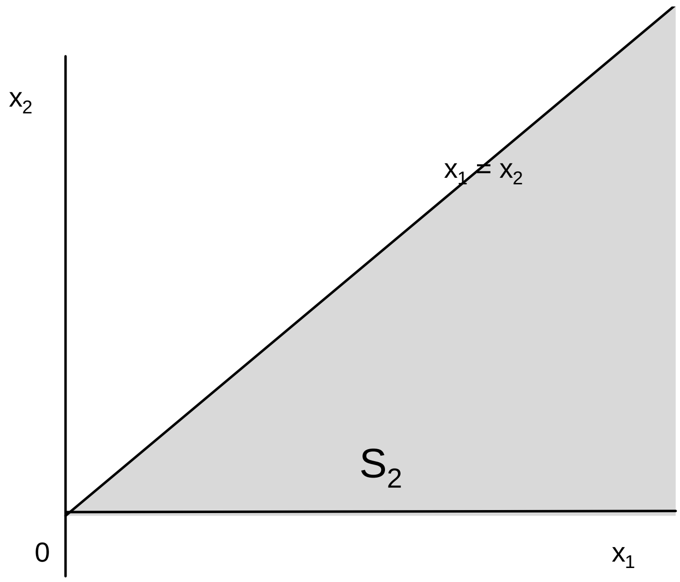

Figure 8.3: $S_2 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 \leq x_2 \leq x_1\}$

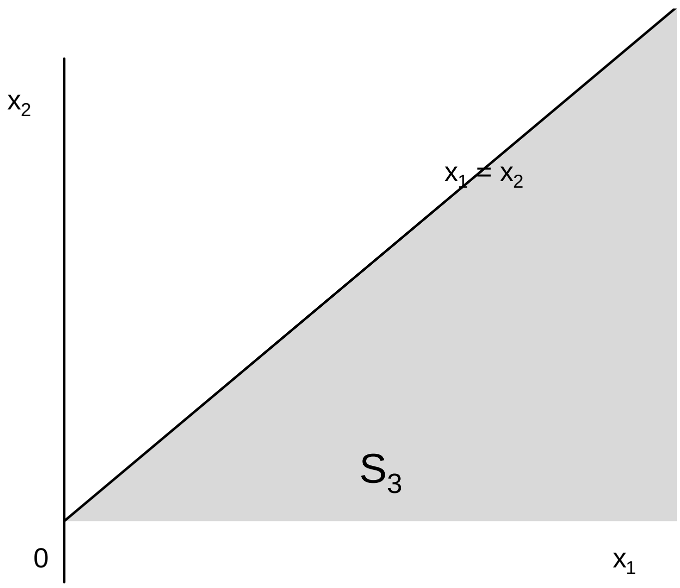

Figure 8.4: $S_3 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 \leq x_1\}$

- A subset S of $\mathcal{R}^n$ is **closed** if **all** the points in the boundary of S lie in S.

For example the boundary of the set $S_1 = \left\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 < x_1\right\}$ in Figure 8.2 is the lines $x_2 = 0$ and $x_2 = x_1$ with $x_1 \geq 0$. These points are not in S_1 so S_1 is open. The boundary of the set $S_2 = \left\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 \leq x_2 \leq x_1\right\}$ in Figure 8.3 is the same as the boundary of S_1 but these points lie within S_2 so S_2 is closed. The lower boundary of the set $S_3 = \left\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 \leq x_1\right\}$ in Figure 8.4 is not in S_3 but the upper boudary is in S_3. The set S_3 is neither open nor closed. You may notice that the set is closed if there all the inquality signs defining the sets are weak $\leq$ and open if all the inquality signs are strong $<$. This is general provided the functions used to define the set are continuous, but has not been proved at this stage.

8.5 Open Sets

8.5.1 Definition

The definition of an open set as a set for which none of the points in the boundary of S lie in S is good for an intuitive understanding of what an open set is. However this definition is not good for proving results about open sets, for which the "official" definition works better. This is:

Definition 53 *A subset S of $\mathcal{R}^n$ is* ***open*** *if for every element $\mathbf{x}$ of S there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S.*

The first result about open sets has to be that the "official" definition is equivalent to the definition in terms of the set boundary. This is:

Proposition 54 *A subset S of $\mathcal{R}^n$ is open if and only if it does not contain any points in its boundary.*

Proof. As S is open then for any element $\mathbf{x}$ of S there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S so does not contain any element of the complement S^C of S. Thus $\mathbf{x}$ cannot be a boundary point of S, so no points in S are in the boundary of S.

Now suppose that S does not contain any of its boundary points, but is not open. As S is not open there is an element $\mathbf{x}$ of S with the property that every open ball $B(\mathbf{x}, \delta)$ contains a point in S^C. As $\mathbf{x}$ is in S this implies that $\mathbf{x}$ is in the boundary of S, contradicting the assumption that S does not contain any of its boundary points. Thus if S does not contain any of its boundary points S must be open. ■

8.5.2 Open Intervals, Open Balls and Open Sets

I have used the term "open interval" frequently. I defined open intervals as sets of the form

$$\begin{aligned}
(a, b) &= \{x : x \in \mathcal{R}, a < x < b\} \\
(a, \infty) &= \{x : x \in \mathcal{R}, a < x\} \\
(-\infty, b) &= \{x : x \in \mathcal{R}, x < b\}.
\end{aligned}$$

The next proposition, which is proved in the appendix to this chapter establishes that this is a sensible form of words.

Proposition 55 *An open interval is an open subset of $\mathcal{R}$.*

The next bit of tidying up of the use of the word open is:

Proposition 56 *An open ball is an open set.*

Again this is proved in the appendix to this chapter.

8.5.3 Unions and Intersections of Open Sets

There are two important results here. The first is

Proposition 57 *The* ***union*** *of a* ***finite or infinite*** *number of open subsets of $\mathcal{R}^n$ is open.*

Proof. If $\mathbf{x}$ is an element of the union it is an element of one of the open sets S, so there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S, and thus a subset of the union of the sets. Thus the union is open. ■

The second result is:

Proposition 58 *The* ***intersection*** *of a* ***finite*** *number of open sets of $\mathcal{R}^n$ is open.*

Proof. Let the subsets be $S_1, S_2....S_n$, and let $\mathbf{x}$ be a point in their intersection. As S_i is open there is a number $\delta_i > 0$ such that the open ball $B(\mathbf{x}, \delta_i)$ is a subset of S_i, that is

$$B(\mathbf{x}, \delta_i) \subset S_i \tag{8.3}$$

Let

$$\delta = \min(\delta_1, \delta_2, ...\delta_n).$$

Thus $0 < \delta \leq \delta_i$ for all i so the open ball $B(\mathbf{x}, \delta)$ is a subset of the open ball $B(\mathbf{x}, \delta_i)$ for all i, that is $B(\mathbf{x}, \delta) \subset B(\mathbf{x}, \delta_i)$. Given relationship 8.3 this implies

$$B(\mathbf{x}, \delta) \subset S_i \text{ for } i = 1, 2...n$$

implying that $B(\mathbf{x}, \delta)$ is a subset of the intersection of $S_1, S_2....S_n$, so the intersection is itself open. ■

Note that the fact that this result applies to a **finite** number of open sets is important. For example that sets $B\left(\mathbf{0}, \frac{1}{n}\right)$ for $n = 1, 2, 3.....$ are open, but their intersection is $\{\mathbf{0}\}$ which is not an open set, because no open ball centred on $\mathbf{0}$ is a subset of $\{\mathbf{0}\}$.

8.6 Closed Sets

8.6.1 Formal Definition

I introduced closed sets as sets that include their boundaries. The "official" definition is

Definition 59 *A set is closed if its complement is open.*

The next proposition is that this definition is equivalent to the definition in terms of the set boundary.

Proposition 60 *A subset S of $\mathcal{R}^n$ is closed if and only if its boundary is a subset of S.*

Proof. Assume the set S is closed so its complement S^C is open. From Proposition 54 as S^C is open none of the points in its boundary lie in S^C, so all the points in the boundary of S^C lie in S. As the boundary of S and the boundary of its complement S^C are the same (Proposition 51) this implies that the boundary of the closed set S lies in S.

Conversely if the boundary of S lies in S there is no point in the boundary of S that lies in S^C. As the boundaries of S and S^C are the same this implies that no point in the boundary of S^C lies in S^C. From Proposition 54 this implies that S^C is an open set so S is a closed set. ■

8.6.2 Closed Sets and Infinite Sequences

You may have seen a different definition of a closed set in terms of limits of infinite sequences. The intuitive idea of the limit of an infinite sequence is that points in the infinite sequence get closer and closer to the limit as you move further and further along the sequence. The formal definition is:

Definition 61 *The infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ in $\mathcal{R}^n$ tends to a limit $\mathbf{x}_L$ in $\mathcal{R}^n$ if for any number $\varepsilon > 0$ there is an integer N such that $\|\mathbf{x}_i - \mathbf{x}_L\| < \varepsilon$, or equivalently $\mathbf{x}_i \in B(\mathbf{x}_L, \varepsilon)$, for all $i > N$.*

The result linking infinite sequences and closed sets is:

Proposition 62 *A subset S of $\mathcal{R}^n$ is closed if and only if, for any infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ whose points are all S that tends to a limit, that limit is in S.*

The proof of this proposition is in the appendix to this chapter. This result shows that I could have defined closed sets in $\mathcal{R}^n$ in terms of infinite sequences, as is sometimes done. However the definition of closed sets as sets whose complements are open is easier to work with.

8.7 Continuous functions

8.7.1 Continuous Functions of a Single Variable

Intuitively a function is continuous if $f(\mathbf{x})$ is very close to $f(\mathbf{x}_0)$ when $\mathbf{x}$ is very close to $\mathbf{x}_0$. Informally a function of a single variable is continuous if you can draw its graph without taking your pen off the paper. The function illustrated in Figure 8.5 is not continuous, the function in Figure 8.6 is continuous. The formal definition of continuity for a function of a single variable defined on an open[1] set is:

[1] At this point I work with a function defined on an open set S because then if $\mathbf{x}$ is in the set any open ball $B(\mathbf{x}, \delta)$ with a small enough δ lies entirely in the set. If the set is not open there are boundary points $\mathbf{x}$ in S for which $B(\mathbf{x}, \delta)$ is not a subset of S for any δ, and the function may not be defined at some points in the open ball.

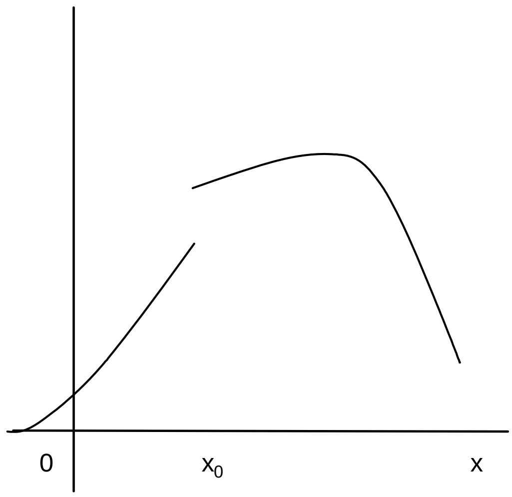

Figure 8.5: A function that is not continuous at x_0

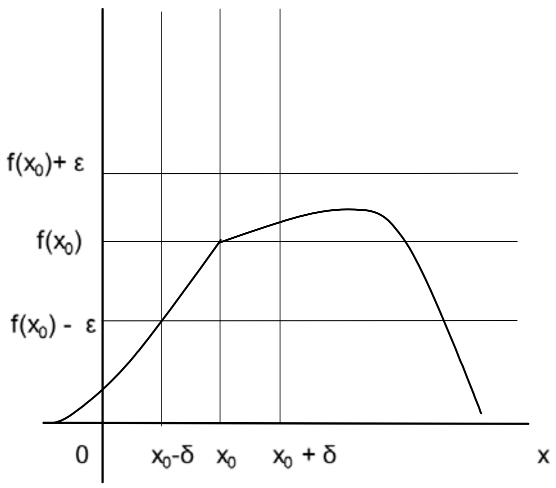

Figure 8.6: A function that is continous at x_0.

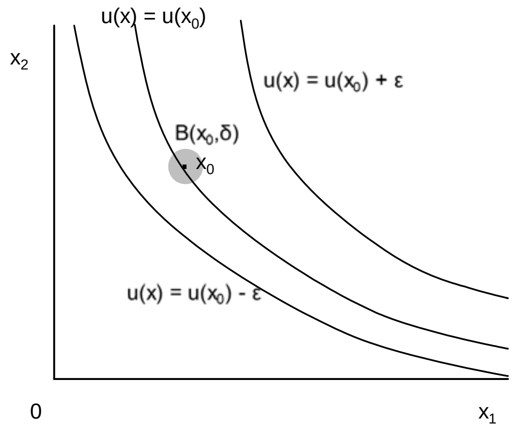

Figure 8.7: Indifference curves for a continuous utility function

Definition 63 *If S is an open subset of $\mathcal{R}$ the function $f : S \to \mathcal{R}$ is* ***continuous*** *if for any $x_0 \in \mathcal{S}$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.*

Figure 8.6 illustrates this formal definition; if x is within a distance δ of x_0 then $f(x)$ is within a distance ε of $f(x_0)$. Note that because the graph of the function in figure 8.6 has a kink at x_0 the function does not have a derivative. Continuity and differentiability are different concetps.

8.7.2 Continuous Functions of Several Variables

The notation for a function of n variables is $f(\mathbf{x})$ where $\mathbf{x}$ is an n vector, that is an element of $\mathcal{R}^n$. The intuitive idea of continuity for functions of several variables is the same as the intuition for functions of a single variable; if $\mathbf{x}$ is very close to $\mathbf{x}_0$ then $f(\mathbf{x})$ is very close to $f(\mathbf{x}_0)$. The formal definition of continuity for functions of several variables requires cutting and pasting the definition for functions of a single variable, replacing $\mathcal{R}$ by $\mathcal{R}^n$ where appropriate, $|x - x_0|$ by $\|\mathbf{x} - \mathbf{x}_0\|$ and $f(x) - f(x_0)$ by $f(\mathbf{x}) - f(\mathbf{x}_0)$.

Definition 64 (Continuity for a Function of Several Variables 1) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is* ***continuous*** *if for any $\mathbf{x}_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\|\mathbf{x} - \mathbf{x}_0\| < \delta$, then $|f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon$.*

It is somewhat easier to visualize with an equivalent definition

Definition 65 (Continuity for a Function of Several Variables 2) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is* ***continuous*** *if for any $\mathbf{x}_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\mathbf{x} \in B(\mathbf{x}_0, \delta)$ then $|f(\mathbf{x}_0) - f(\mathbf{x})| < \delta$*

To see that the two definitions are equivalent observe that the set of values of $\mathbf{x}$ for which $\|\mathbf{x}-\mathbf{x}_0\|<\delta$ is the open ball $B(\mathbf{x}_0,\delta)$.

Figure 8.7 illustrates this definition for one of economists' favorite functions and diagrams, a utility function and indifference curves. There is an open ball $B(\mathbf{x}_0,\delta)$ that lies in the set between the two indifference curves with $u(\mathbf{x})=u(\mathbf{x}_0)-\varepsilon$ and $u(\mathbf{x})=u(\mathbf{x}_0)-\varepsilon$. For all points in this open ball $|u(\mathbf{x})-u(\mathbf{x}_0)|<\varepsilon$.

8.8 Closed Sets, Bounded Sets, Compact Sets and Continuous Functions

For economists one of the most important results on continuous functions is:

Theorem 66 *If S is a* ***closed and bounded*** *subset of $\mathcal{R}^n$ and the function $f:S\to\mathcal{R}$ is* ***continuous*** *the function* ***has a maximum and a minimum*** *on S, that is there are elements $\mathbf{x}_{\min}$ and $\mathbf{x}_{\max}$ of S with the property that*

$$f(\mathbf{x}_{\min})\le f(\mathbf{x})\le f(\mathbf{x}_{\max})$$

for all $\mathbf{x}$ in S.

The theorem is important due to economists' obsession with maximization as it gives conditions under which you can be sure that the function has a maximum and minimum. The proof is somewhat complicated and I do not give it here.

This definition is no use without a definition of a bounded set:

Definition 67 *A subset S of $\mathcal{R}^n$ is* ***bounded*** *if there is an element y of $\mathcal{R}$ with the property that*

$$\|\mathbf{x}\|<y \text{ for all } \mathbf{x} \text{ in } S.$$

Another way of saying this is that the distance between $\mathbf{x}$ and $\mathbf{0}$ is less than y for all elements of S. If $n=1$ this simply means that $-y<x<y$ for all x in S. If $n=2$ this means that S lies inside a circle with radius y centered on the origin. More generally it means that all $\mathbf{x}$ in S lie in the open ball $B(\mathbf{0},y)$.

It is important that S be both closed and bounded. For example the interval $(0,1)$ is bounded but not closed, the function $f(x)=x^{-1}$ is continuous on $(0,1)$ but has no maximum, as $f(x)$ grows without limit as x gets closer and closer to 0. The interval $[0,\infty)$ is closed but not bounded, and the continuous function $f(x)=x^2$ grows without limit as x tends to infinity.

If you have done some general topology you will have come across the term "compact". There is a general definition of a compact set and then a theorem (the Heine-Borel Theorem) which states that the sets in $\mathcal{R}^n$ that are compact are the sets that are closed and bounded, so when working in $\mathcal{R}^n$ you should interpret compact as meaning closed and bounded. If you have never done any general topology ignore this comment.

8.9 Continuity for Consumer Theory

8.9.1 The Definition

Continuity is important at one point in consumer theory, showing that the continuity assumption on preferences implies the existence of a continuous utility function that represents the preferences. In order to understand this result you need to know that continuity can be defined in the following way.

Definition 68 (Continuity for a Function of Several Variables 3) *If S is a subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

For this to make sense this definition has to be equivalent to Definitions 64 and 65 of continuous functions. The rest of the chapter demonstrates that this is indeed so. This is very much an exercise in pure mathematics with a number of real analysis type arguments with ε and δ. I suggest skipping this section if you have never seen any real analysis.

8.9.2 Level Sets, Upper Contour Sets, Lower Contour Sets

It is useful to have some terminology.

Definition 69 *If S is a subset of $\mathcal{R}$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) = y\}$$

*of elements of S for which $f(\mathbf{x}) = y$ is called a **level set**.*

This may be a new piece of vocabulary, but economists are very familiar with level sets; we call the level set of a utility function an indifference curve, and the level set of a production function an isoquant.

Definition 70 *If S is a subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$$

*of elements of S for which $f(\mathbf{x})$ is greater than y is called an **upper contour set**.*

Definition 71 *If S is a subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$$

*of elements of S for which $f(\mathbf{x})$ is less than y is called a **lower contour set**.*

Figure 8.8 illustrates these sets for a utility function.

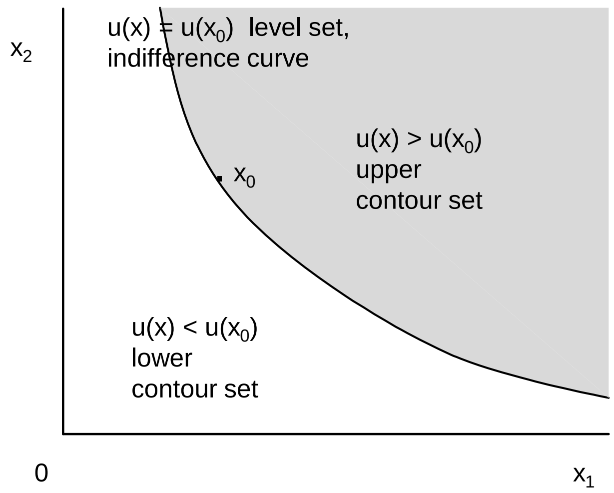

Figure 8.8: Level set, upper contour set and lower contour set for a utility function

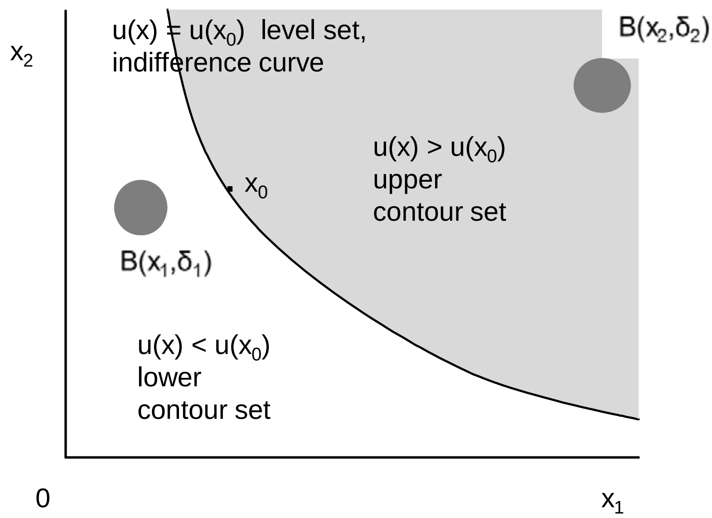

Figure 8.9: Open upper and lower contour sets

8.9.3 Open Set and Closed Set Definition of Continuity

The argument that Definition 68 of continuity is equivalent to Definitions 64 and 65 starts from the following proposition.

Proposition 72 *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is continuous if and only if for all elements y of $\mathcal{R}$ the upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$ and the lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$ are both open.*

Figure 8.9 illustrates the conditions of this thereon for a utility function. As the lower contour set is open, for any point $\mathbf{x}_1$ in the lower contour set, there is an open ball $B(\mathbf{x}_1, \delta_1)$ that is a subset of the lower contour set. As the upper contour set is open, for any point $\mathbf{x}_2$ in the upper contour set, there is an open ball $B(\mathbf{x}_2, \delta_2)$ that is a subset of the upper contour set.

The proof of Proposition 72 is suggested by Figures 8.8 and 8.9.

Proof. The function $f : S \to \mathcal{R}$ is continuous if for any $x_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\mathbf{x} \in B(\mathbf{x}_0, \delta)$ then $|f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon$. In particular if $f(\mathbf{x}_1) < y$ so $y - f(\mathbf{x}_1) > 0$ there is an open ball $B(\mathbf{x}_1, \delta_1)$ on which $|f(\mathbf{x}) - f(\mathbf{x}_1)| < y - f(\mathbf{x}_1)$. As $f(\mathbf{x}) - f(\mathbf{x}_1) \leq |f(\mathbf{x}) - f(\mathbf{x}_1)|$ which implies that $f(\mathbf{x}) - f(\mathbf{x}_1) < y - f(\mathbf{x}_1)$ so $f(\mathbf{x}) < y$. Thus the lower contour set is open. Similarly if $f(\mathbf{x}_2) > y$ so $f(\mathbf{x}_2) - y > 0$ there is an open ball $B(\mathbf{x}_2, \delta_2)$ on which $|f(\mathbf{x}_2) - f(\mathbf{x})| < f(\mathbf{x}_2) - y$. As $f(\mathbf{x}_2) - f(\mathbf{x}) \leq |f(\mathbf{x}_2) - f(\mathbf{x})|$ this implies that $f(\mathbf{x}_2) - f(\mathbf{x}) < f(\mathbf{x}_2) - y$ so $y < f(\mathbf{x})$. Thus the upper contour set is open. Hence the upper and lower contour sets are open if the function is continuous.

To prove the converse result note that if the upper contour set

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}_0) - \varepsilon < f(\mathbf{x})\}$$

and the lower contour set

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < f(\mathbf{x}_0) + \varepsilon\}$$

are open then, from Proposition 58, their intersection is also open. This intersection is the set on which $f(\mathbf{x}_0) - \varepsilon < f(\mathbf{x}) < f(\mathbf{x}_0) + \varepsilon$, or equivalently the set on which $-\varepsilon < f(\mathbf{x}) - f(\mathbf{x}_0) < \varepsilon$, that is the set $\{\mathbf{x} : \mathbf{x} \in S,\ |f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon\}$. The point $\mathbf{x}_0$ is an element of this set, hence as the set is open there is an open ball $B(\mathbf{x}_0, \delta)$ with the property that all elements of the open ball lie in the set $\{\mathbf{x} : \mathbf{x} \in S,\ |f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon\}$. This is what is needed to establish continuity. ■

This proposition makes possible yet another equivalent definition of continuity.

Definition 73 (Continuity for a Function of Several Variables 4) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$ and the lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$ are both open.*

Now suppose that $f(\mathbf{x}) : S \to \mathcal{R}$ is continuous and consider the complement in S of the open upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$, that is the set

$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$. This is by definition a closed set[2] . Similarly the complement in S of the open lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$, that is the closed set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$. Given Definition 73 I can give yet another definition of continuity.

Definition 74 (Continuity for a Function of Several Variables 5) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

I have been assuming throughout this argument that S is an open set. With a little more mathematical sophistication[3] I can drop the requirement that S is an open set. This is useful thing to do, because for may purposes it is necessary to think of functions defined on $\mathcal{R}^{n+}$ which is defined as the set of vectors with $x_i \geq 0$ for $i = 1, 2...n$. The set $\mathcal{R}^{n+}$ is closed.

Definition 75 (Continuity for a Function of Several Variables) *If S is a subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

This is the definition used in consumer theory.

8.10 Appendix: Proofs

8.10.1 Proof That An Open Interval is an 0pen Subset of $\mathcal{R}$

If $a < b$, $a < x < b$ and $\varepsilon = \min(x - a, b - x)$ then as $a < x < b$

$$\varepsilon > 0. \tag{8.4}$$

As $\varepsilon = \min(x - a, b - x) \leq x - a$ and $\varepsilon > 0$

$$a \leq x - \varepsilon < x - \frac{1}{2}\varepsilon < x$$

so

$$a < x - \frac{1}{2}\varepsilon < x < b. \tag{8.5}$$

As $\varepsilon = \min(x - a, b - x) \leq b - x$

$$b \geq x + \varepsilon > x + \frac{1}{2}\varepsilon > x > a$$

[2] I am being somewhat sloppy with definitions of open and closed sets here. If you have not noticed my sloppiness ignore this footnote. The difficulty is that I am assuming that S is open and f is continuous, in which case the set $\{\mathbf{x} : \mathbf{x} \in S, f(\mathbf{x}) \leq y\}$ that I am calling the complement of the upper contour set in S is not a closed subset of $\mathcal{R}^n$ unless $S = \mathcal{R}^n$. I can get away with this by defining an open set in S as the intersection of S and a set that is open in $\mathcal{R}^n$, and a closed set in S as the complement in S of an open set in S. If S is open the open sets in S are open in $\mathcal{R}^n$, but the closed sets in S are not closed in $\mathcal{R}^n$. If S is closed some of the open sets in S are not open in $\mathcal{R}^n$, but the closed sets in S are closed in $\mathcal{R}^n$. If you know about general topology you will recognize that either way I am setting up a topology on S, and know that this makes mathematical sense.

[3] In order to do this I have to define an open set in S as the intersection of an open set in $\mathcal{R}^n$, and then define a closed subset of S as the complement in S of an open subset in S. See footnote 2.

so

$$a < x < x + \frac{1}{2}\varepsilon. \tag{8.6}$$

Taken together inequalities 8.4, 8.5 and 8.6 imply that

$$a < x - \frac{1}{2}\varepsilon < x < x + \frac{1}{2}\varepsilon < b. \tag{8.7}$$

The set of points satisfying $x - \frac{1}{2}\varepsilon < x < x + \frac{1}{2}\varepsilon$ is the open ball $B\left(x,\frac{\varepsilon}{2}\right)$. Inequality 8.7 then implies that $B\left(x,\frac{\varepsilon}{2}\right)$ is a subset of the intervals (a, b) (a, ∞) and $(-\infty, b)$. Thus these intervals are open.

8.10.2 Proof That An Open Ball is an Open Subset of $\mathcal{R}^n$

The open ball $B(\mathbf{x}_0, \delta)$ is the set of points satisfying $\|\mathbf{x} - \mathbf{x}_0\| < \delta$. Note that if $\mathbf{x}$ is an element of $B(\mathbf{x}_0, \delta)$ then $\delta - \|\mathbf{x} - \mathbf{x}_0\| > 0$. Let $\mathbf{z}$ be an element of the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$. Then

$$\|\mathbf{z} - \mathbf{x}\| < \delta - \|\mathbf{x} - \mathbf{x}_0\|.$$

From the triangle inequality (proved in the previous chapter on vectors)

$$\|\mathbf{z} - \mathbf{x}_0\| = \|\mathbf{z} - \mathbf{x} + \mathbf{x} - \mathbf{x}_0\| \leq \|\mathbf{z} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\|.$$

The last two inequalities imply that

$$\|\mathbf{z} - \mathbf{x}_0\| \leq \|\mathbf{z} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < \delta - \|\mathbf{x} - \mathbf{x}_0\| + \|\mathbf{x} - \mathbf{x}_0\| = \delta.$$

Thus if $\mathbf{z}$ is an element of the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$ then $\|\mathbf{z} - \mathbf{x}_0\| < \delta$, so $\mathbf{z}$ is an element of the open ball $B(\mathbf{x}_0, \delta)$. Thus the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$ is a subset of $B(\mathbf{x}_0, \delta)$, so the open ball $B(\mathbf{x}_0, \delta)$ is an open set.

8.10.3 Infinite Sequences and Closed Sets

The result here is

Proposition 76 *A set S is closed if and only if for any infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ whose points are all S that tends to a limit, that limit is in S.*

Proof. Suppose that S is closed and that $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ is an infinite sequence of points in S that tend to a limit in S^C. As S is closed S^C is open so there is an open ball $B(\mathbf{x}_L, \varepsilon)$ that is a subset of S^C. As $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ tends to $\mathbf{x}_L$ the point $\mathbf{x}_i$ is an element of $B(\mathbf{x}_L, \varepsilon)$ and thus of S^C for all sufficiently large i. But by assumption all points in the sequence lie in S so cannot be in S^C. This contradiction implies that the limit of infinite sequence of points a closed set S must lie in S.

Now suppose that the limit of any infinite sequence of points in S lies in S, but S is not closed. As S is not closed, S^C is not open. This implies that there is a point $\mathbf{x}$ in S^C for which every open ball $B(\mathbf{x}, \varepsilon)$ contains a point that is not in S^C and therefore is in S, regardless of how small ε is. Thus there is an infinite sequence of points $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ that are not in S^C for which $\mathbf{x}_i \in B\left(\mathbf{x}, \frac{1}{i}\right)$. This infinite sequence tends to $\mathbf{x}$. As none of the points in the infinite sequence are in S^C they are all in S. However the limit is in S^C. This contradicts the supposition on limits of infinite sequences and S not being closed that I started with. Thus S is closed. ■